Technical Report
Cloud Resume Challenge – Serverless GraphQL Application

1. Project Overview
This project is an implementation of the Cloud Resume Challenge designed to demonstrate practical skills in cloud computing, serverless architecture, and modern web development. The system consists of a cloud-hosted frontend, a serverless GraphQL backend, and a database layer.

2. Motivation and Problem Statement
The goal of this project was to design a real-world cloud-native application that demonstrates backend engineering, API development, and deployment on cloud infrastructure beyond static resume hosting.

3. System Architecture
The architecture follows a serverless approach with a React-based frontend hosted on Vercel, a GraphQL backend using Apollo Server deployed on AWS Lambda via API Gateway, and MongoDB as the database.

4. Technologies Used
Frontend: React, Tailwind CSS, Vercel
Backend: Node.js, Apollo GraphQL, AWS Lambda, API Gateway
Database: MongoDB
Testing: Mocha, Chai (TDD)

5. Backend Design
The backend exposes GraphQL queries and mutations to handle visitor count and data retrieval. Apollo Server enables efficient schema-based data access and integrates seamlessly with AWS Lambda.

6. Test-Driven Development
The backend was developed using Test-Driven Development principles. Unit tests were written using Mocha and Chai to validate resolver logic, database operations, and error handling.

7. Security
Security best practices include least-privilege IAM roles, environment variable configuration, and HTTPS communication between services.

8. Deployment
The frontend is deployed using Vercel with continuous deployment. The backend scales automatically using AWS Lambda and API Gateway, ensuring high availability and low operational cost.

9. Challenges
Key challenges included managing database connections in a serverless environment and testing Lambda functions. These were resolved through connection reuse and isolated unit testing.

10. Conclusion

This project demonstrates strong understanding of cloud-native application design, backend systems, and serverless computing. It reflects readiness for advanced academic study in Information Technology.